Rajsekhar Mohapatra

Theoretical and Numerical Astrophysics

Humboldt Fellow, AIP Potsdam

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Research interests

My research combines theoretical modelling and numerical fluid dynamics to investigate turbulence, magnetic fields, cooling and feedback across the interstellar medium, galactic atmospheres and galaxy clusters. I use high-resolution simulations to connect the underlying gas physics to observations and models for larger-scale simulations.

Academic and research appointments

Oct 2026–present	**Humboldt Fellow** Leibniz Institute for Astrophysics Potsdam (AIP), Germany.
Sep 2022–Sep 2026	**Postdoctoral Research Associate** Department of Astrophysical Sciences, Princeton University, USA.

Education

Oct 2018–Sep 2022	**PhD in Astronomy**, Australian National University, Australia. Thesis: *The Role of Stratification, Turbulence and Cooling in the Intracluster Medium*. Supervisor: Prof. Christoph Federrath.
Aug 2014–Jun 2018	**BSc in Physics**, Indian Institute of Science, Bengaluru, India. Thesis: *Studying Turbulence in the Intracluster Medium*. Supervisor: Prof. Prateek Sharma.

Awards and scholarships

From Oct 2026	**Humboldt Fellowship**, Alexander von Humboldt Foundation, Germany.
2018–2022	**University Research Scholarship**, Australian National University.
2019, 2022	**International Travel Grant**, Research School of Astronomy and Astrophysics, Australian National University.
2013–2018	**Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship**, Department of Science and Technology, India.

Publications

First-author papers

- **Mohapatra, R.**, Dutta, A., and Sharma, P. "Multiphase gas in Circumgalactic cloud complexes: Insights from kiloparsec-scale Magnetohydrodynamic Turbulence Simulations". In: *The Open Journal of Astrophysics* 9 (2026), p. 64831. DOI: 10.33232/001c.164831.

- **Mohapatra**, **R.**, Lancaster, L., Fielding, D., Quataert, E., and Bryan, G. L. "How Magnetic Fields Regulate Cooling and Mixing in Turbulent Radiative Mixing Layers". In: (2026). arXiv: 2609.29782 [astro-ph.GA] **(submitted)**.
- **Mohapatra**, **R.**, Quataert, E., Fielding, D., and Guo, M. "The Type Ia Supernova and Asymptotic Giant Branch Stellar Ejecta-regulated Interstellar Medium of Massive Galaxies". In: *ApJ* 989, 103 (2025). DOI: 10.3847/1538-4357/adea6c.
- **Mohapatra**, **R.** and Quataert, E. "Multiphase Gas in Elliptical Galaxies: The Role of Type Ia Supernovae". In: *ApJ* 965, 105 (2024). DOI: 10.3847/1538-4357/ad2940.
- **Mohapatra**, **R.**, Sharma, P., Federrath, C., and Quataert, E. "Multiphase condensation in cluster haloes: interplay of cooling, buoyancy, and mixing". In: *MNRAS* 525 (2023), pp. 3831–3848. DOI: 10.1093/mnras/stad2574.
- **Mohapatra**, **R.**, Federrath, C., and Sharma, P. "Multiphase turbulence in galactic haloes: effect of the driving". In: *MNRAS* 514 (2022), pp. 3139–3159. DOI: 10.1093/mnras/stac1610.
- **Mohapatra**, **R.**, Jetti, M., Sharma, P., and Federrath, C. "Characterizing the turbulent multiphase haloes with periodic box simulations". In: *MNRAS* 510 (2022), pp. 3778–3793. DOI: 10.1093/mnras/stab3603.
- **Mohapatra**, **R.**, Jetti, M., Sharma, P., and Federrath, C. "Velocity structure functions in multiphase turbulence: interpreting kinematics of Hα filaments in cool-core clusters". In: *MNRAS* 510 (2022), pp. 2327–2343. DOI: 10.1093/mnras/stab3429.
- **Mohapatra**, **R.**, Federrath, C., and Sharma, P. "Turbulent density and pressure fluctuations in the stratified intracluster medium". In: *MNRAS* 500 (2021), pp. 5072–5087. DOI: 10.1093/mnras/staa3564.
- **Mohapatra**, **R.**, Federrath, C., and Sharma, P. "Turbulence in stratified atmospheres: implications for the intracluster medium". In: *MNRAS* 493 (2020), pp. 5838–5853. DOI: 10.1093/mnras/staa711.
- **Mohapatra**, **R.** and Sharma, P. "Turbulence in the intracluster medium: simulations, observables, and thermodynamics". In: *MNRAS* 484 (2019), pp. 4881–4896. DOI: 10.1093/mnras/stz328.

Second- and third-author papers

- Desire, T., Kim, C.-G., and **Mohapatra**, **R.** "A Comparative Study of Isothermal Turbulence Statistics: Fourier Space Driving vs. Point Source Driving". In: (2026). DOI: 10.48550/arXiv.2606.14946. arXiv: 2606.14946 [astro-ph.GA] **(submitted)**.
- Lancaster, L., Fielding, D., **Mohapatra**, **R.**, and Bryan, G. L. "The Origin of Da Scaling: Suppressed Cooling in Fast-cooling Mixing Layers". In: *ApJ* 1008, 13 (2026). DOI: 10.3847/1538-4357/ae908e.
- Lancaster, L., **Mohapatra**, **R.**, Fielding, D. B., and Bryan, G. L. "Ceci n'est pas une Couche de Mélange: The Meaning of Resolved Turbulent Radiative Mixing". In: *ApJ* 1005, 222 (2026). DOI: 10.3847/1538-4357/ae7714.
- Gatuzz, E., **Mohapatra**, **R.**, Federrath, C., Sanders, J. S., Liu, A., Walker, S. A., and Pinto, C. "Measuring the hot ICM velocity structure function using XMM-Newton observations". In: *MNRAS* 524 (2023), pp. 2945–2953. DOI: 10.1093/mnras/stad2039.

Other co-authored papers

- Fournier, M., Grete, P., Brüggen, M., O'Shea, B. W., Prasad, D., Wibking, B. D., Glines, F. W., and **Mohapatra**, **R.** "XMAGNET: Velocity structure functions of active galactic nucleus-driven turbulence in the multiphase intracluster medium". In: *A&A* 698, A121 (2025). DOI: 10.1051/0004-6361/202554278.

- Gatuzz, E., Sanders, J., Liu, A., Fabian, A., Pinto, C., Russell, H., Eckert, D., Walker, S., ZuHone, J., and **Mohapatra**, **R.** "Chemical enrichment of the intracluster medium within the A3266 cluster: I. Radial profiles". In: *A&A* 698, A94 (2025). DOI: 10.1051/0004-6361/202453360.
- Gatuzz, E., Sanders, J., Liu, A., Fabian, A., Pinto, C., Russell, H., Eckert, D., Walker, S., ZuHone, J., and **Mohapatra**, **R.** "Measuring the intracluster medium velocity structure within the A3266 galaxy cluster". In: *A&A* 692, A108 (2024). DOI: 10.1051/0004-6361/202451640.

Competitive computing allocations

ACCESS credits are allocations redeemable for computing time across US national supercomputers. DeltaAI equivalents below use the current conversion of 7,500 H100 GPU-hours per million credits.

2026	**ACCESS — Co-principal investigator** 10 million ACCESS credits; equivalent to 75,000 DeltaAI H100 GPU-hours.
2025	**Frontier — Co-investigator** 800,000 node-hours, Oak Ridge National Laboratory, USA.
2025	**ACCESS — Co-principal investigator** 20 million ACCESS credits; equivalent to 150,000 DeltaAI H100 GPU-hours.
2024	**ACCESS — Co-principal investigator** 13 million ACCESS credits; equivalent to 97,500 DeltaAI H100 GPU-hours.
2023	**ACCESS — Principal investigator** 3 million ACCESS credits (45,000 Delta A100 GPU-hours).

Research software development and numerical expertise

Methods: High-resolution hydrodynamic and magnetohydrodynamic simulations; driven and stratified turbulence; radiative cooling; resolved supernova heating; turbulent mixing layers; radiative shocks.

Simulation codes: AthenaK, FLASH and PLUTO.

AthenaK research development: I maintain athenak-RM, my public research fork of AthenaK, with a turbulence driver and on-the-fly analysis tools.

Simulation analysis software: I develop AthenaResearchRM, an open-source Python package for simulation analysis and visualisation, with CPU/GPU support and MPI parallelism.

Programming and analysis: C, C++, Fortran, Python.

Computing facilities: USA: Frontier, Delta and DeltaAI; Princeton's Della and Stellar; Australia: Gadi and Raijin; Finland/EuroHPC: LUMI.

Mentoring and supervision

2026	Co-supervision and mentoring of Xiaoqi Jiang, a master's student at the University of Chicago, USA, during a summer research project.
2026	Supervision and mentoring of Tanya Zhou, undergraduate summer student, Princeton University, USA.
2025–2026	Supervision and mentoring of Prisha Sahu, high-school student, Texas, USA.
2024–2026	Co-supervision and mentoring of Tejahni Desire, bachelor's thesis student from Princeton University, USA; student-led paper submitted.

2023–2026 Mentor, graduate student–postdoc mentoring programme, Princeton University.

2025 Supervision and mentoring of Yash Padaya, a master's student at Liverpool John Moores University, UK, during a summer research project.

2021–2022 Co-supervision and mentoring of Mrinal Jetti, master's thesis student from IIT Madras, Chennai, India; two papers published in MNRAS with the student as a co-author (paper 1, paper 2).

2019–2021 Supervision and mentoring of two high-school students through Canberra's MSATT program, Canberra, Australia.

Teaching

Jan 2020–May 2021 **Tutor in Physics**, Australian National University.
Physics of Materials (2020); Advanced Electromagnetism (2021).

2021 **Tutor, Indigenous Tuition Program**, Tjabal Indigenous Higher Education Centre, Australian National University. Provided academic tutoring for Indigenous Australian students.

2020 **Guest lecturer, Astrophysical Processes (ASTR6013)**, Australian National University. Guest lectures in the graduate course convened by Prof. Christoph Federrath.

Academic service

2022–present **Journal refereeing:** Astronomy & Astrophysics; The Astrophysical Journal; Astronomy and Computing; Monthly Notices of the Royal Astronomical Society; The Open Journal of Astrophysics; Frontiers.

2024–2026 **Postdoctoral representative, Departmental Climate Committee**, Department of Astrophysical Sciences, Princeton University.

2024 **Proposal referee**, Hubble Space Telescope (HST) Cycle 32.

2024 **Workshop organising committee**, *Synergistic Approaches to Particle Transport in Magnetized Turbulence*, Princeton Center for Theoretical Science, Princeton University.

2022 **Volunteer, IAU General Assembly**, Busan, South Korea.

2020–2022 **Student representative, Inclusion, Diversity, Equity and Access Committee**, Research School of Astronomy & Astrophysics (RSAA), Australian National University.

2019 **Organiser, Mount Stromlo Student Seminars**, Australian National University.

Outreach and public engagement

2023–2025 **Public observing team**, Department of Astrophysical Sciences, Princeton University.

2019–2022	**Outreach Assistant**, Mount Stromlo Observatory, Australian National University. Organised more than 50 school visits for students in Years 3–10, spanning approximately 8–16 years of age.
	Rural school outreach: Visited schools in rural Australia to deliver astronomy outreach activities, New South Wales, Australia.
	Public talks: Delivered astronomy talks to audiences of more than 100 people, Mount Stromlo Observatory, Australia.
	Observatory open days: Organised ANU Open Day tours at Siding Spring Observatory, Australia.
2020	**Online public talks:** Delivered live astronomy talks on YouTube and Facebook during the COVID-19 pandemic.
2015–2018	**IISc open days:** Organised Open Day tours at the Indian Institute of Science, Bengaluru, India.

Seminars

Invited seminars

Apr 2026	*KICP Seminar*, University of Chicago, USA.
Mar 2026	University of Massachusetts Amherst, USA.
Feb 2026	Australian National University, Canberra, Australia.
Mar 2024	*Galaxy Talk*, Center for Computational Astrophysics, Flatiron Institute, New York, USA.
Apr 2023	*Monday Science Seminar*, Department of Astronomy, University of Wisconsin–Madison, USA.

Other seminars

Sep 2026	*Oxford Simulators Meeting*, University of Oxford, UK.
Apr 2026	*CIERA Seminar*, Northwestern University, Evanston, USA.
Feb 2026	*Seminar*, Swinburne University of Technology, Melbourne, Australia.
Feb 2026	University of Western Australia, Perth, Australia.
Dec 2025	Cardiff University, UK.
Dec 2025	*Plasma group meeting*, University of Oxford, UK.
Nov 2025	*Cosmology Lunch*, Max Planck Institute for Astrophysics, Garching, Germany.
May 2025	Yonsei University, Seoul, South Korea.
Nov 2024	*Galaxy Clusters Meeting*, Center for Astrophysics, Harvard University, Cambridge, USA.
Oct 2024	*Cosmology Lunch*, Max Planck Institute for Astrophysics, Garching, Germany.
Oct 2024	*Galaxy Lunch*, Leiden University, Netherlands.
Sep 2024	*Galaxy Coffee*, Max Planck Institute for Astronomy, Heidelberg, Germany.
May 2024	*Galaxy Discussion*, Institute of Astronomy, University of Cambridge, UK.
May 2024	*Galaxy Lunch Talk*, Department of Physics, University of Oxford, UK.

Feb 2024	*KIPAC XOC Group Talk*, Stanford University, USA.
Feb 2024	*Tea Talk*, Caltech, Pasadena, USA.
Apr 2023	*Galaxy Lunch Seminar*, Department of Astronomy, Yale University, New Haven, USA.

Conferences, workshops and research programmes

Invited talks

Jul 2026	*8th ICM Theory and Computation Workshop*, Brno, Czech Republic.
Jun 2024	*7th ICM Theory and Computation Workshop*, University of Michigan, Ann Arbor, USA.

Contributed talks

Jul 2026	*The Many Scales of Galaxy Environments (GalScale26)*, Ascona, Switzerland.
Nov 2025	*Modelling of Multiphase Astrophysical Media*, Schloss Ringberg, Germany.
Jul 2025	*National Astronomy Meeting*, Durham, UK.
Dec 2024	*Baryons Beyond Galactic Boundaries*, IUCAA, Pune, India.
Aug–Sep 2024	*Toward a Holistic Understanding of the Multi-scale, Multiphase Circumgalactic Medium*, Aspen Center for Physics, USA.
Feb 2024	*Turbulence in the Universe*, KITP conference, Santa Barbara, USA.
Jan–Mar 2024	*Turbulence in Astrophysical Environments*, KITP research programme, Santa Barbara, USA.
Feb 2023	*Oasis in the Cosmic Desert*, Arizona State University, USA.
Nov 2022	*Exploring the Hot and Energetic Universe*, Barcelona, Spain.
Aug 2022	*XXXI IAU General Assembly*, BEXCO, Busan, South Korea.